

THE GARDEN OF FORKING PATHS

Multiverse
Analysis ·
Specification
Curve Analysis ·
Vibration of
Effects

At a glance: These three tools share a common purpose: making visible how much a published finding depends on analytic choices the researcher made — choices that are rarely disclosed, often arbitrary, and collectively capable of turning a null result into a significant one, or vice versa.

1. The Shared Problem: Analytical Fragility

Every empirical study involves dozens of decisions that are never fully pre-specified: which cases to exclude, which covariates to control for, how to operationalize the key variables, which model family to use, what to do with outliers. Taken individually, each choice seems reasonable. But taken collectively they form what Andrew Gelman and Eric Loken described in 2014 as a 'garden of forking paths': a branching decision tree where the researcher, consciously or not, tends to follow the path that yields significance.

The problem is not that researchers are dishonest. It is that the garden is invisible in the published paper. Readers see one path — the one that reached significance — with no indication of how many equally valid paths were never reported, or what they would have shown. This selective presentation inflates the false-positive rate in the literature without any individual act of fraud.

The three tools described in this guide all operate by making the rest of the garden visible. They differ in how they do this, what data they require, and what quantity they ultimately estimate.

2. At-a-Glance Comparison

Core question	Which results are possible across all defensible specifications? What is the distribution across all valid specs with inferential tests? How sensitive is the effect to covariate choice?
Primary output	Distribution of effect sizes across all specification combinations
Disciplines	Psychology, sociology, political science; economics (SCA); epidemiology (VoE)
Raw data needed?	Yes — all three require individual-level data to re-run analyses from scratch
Introduced	Multiverse: Steegen et al. 2016 · Specification Curve: Simonsohn et al. 2020 · Vibration of Effects: Patel et al. 2015

Tool A · Multiverse Analysis

Running all defensible specifications and displaying the full distribution of results

Overview

Multiverse analysis, introduced by Sara Steegen and colleagues in 2016, takes the position that there is rarely a single correct specification of a statistical model — there is instead a set of equally defensible choices. Rather than choosing one and reporting it as 'the' result, multiverse analysis runs all of them and presents the entire distribution of effect sizes and significance decisions as the finding.

Methodological Overview

The analyst begins by enumerating every analytic decision point where multiple choices would be scientifically defensible. Common decision nodes include:

- Outlier exclusion rules (e.g., ± 2 SD vs. ± 3 SD vs. no exclusion)
- Variable operationalization (e.g., scale total vs. mean; raw vs. log-transformed)
- Covariate inclusion (e.g., include age, exclude age, include age and gender)
- Subsample restrictions (e.g., full sample vs. participants meeting an attention check)
- Model family (e.g., OLS vs. robust regression vs. mixed effects)

Each unique combination of one choice at each node constitutes a 'specification'. If there are 3 outlier rules, 2 operationalization, 3 covariate sets, and 2 subsamples, there are $3 \times 2 \times 3 \times 2 = 36$ specifications. The

multiverse runs all 36 and records the effect size and p-value for each. Results are displayed as a two-panel figure: effect size estimates sorted by magnitude (top panel), and a grid showing which analytic choices produced which estimates (bottom panel).

Example Application

Oswald et al. conducted a multiverse-style re-analysis of studies on whether implicit racial bias (IAT scores) predicted discriminatory behavior. Using different inclusion criteria, moderator sets, and outcome operationalizations, the distribution of effect sizes across specifications ranged from near-zero to substantial — with the original published estimate sitting at the positive tail of the plausible range, not the center. The multiverse made visible that the headline estimate was among the most favorable of all defensible results.

Requirements

Raw data	Required. The multiverse re-runs the analysis from scratch under each specification. Published summary statistics alone are not sufficient.
Code access	Highly beneficial. Reproducible analysis code (R, Python, Stata) allows automated re-running. Without code, each specification must be run manually.
Decision enumeration	The researcher must explicitly specify all choice nodes and defensible options at each. This requires domain knowledge — there is no automated way to enumerate valid choices.
R package	The multiverse R package (Steegeen et al.) automates specification enumeration and running. <code>specr</code> is an alternative with a tidy interface.
Interpretive rule	Results require a decision rule for 'robust': proportion of significant specs, sign consistency, or the width of the effect size distribution across the full multiverse.

Tool B · Specification Curve Analysis

Plotting all valid specifications with inferential permutation tests across the full curve

Overview

Specification curve analysis (SCA), developed by Uri Simonsohn, Joseph Simmons, and Leif Nelson, takes a similar approach to multiverse analysis but adds a formal inferential layer: instead of just displaying the distribution, it tests whether the observed curve is more extreme than would be expected if the null hypothesis were true across all specifications.

Where multiverse analysis asks 'what do the results look like across the choice space?', SCA asks 'is there genuine signal in the data, or is the pattern of results consistent with chance given the number of choices made?' The specification curve is simultaneously a visualization tool and a statistical test.

Methodological Overview

SCA proceeds in three steps. First, define the universe of specifications — all combinations of analytic choices that are (a) not clearly wrong and (b) at least defensible given the research question. Second, estimate and plot the specification curve: each specification is run, its effect and significance recorded, results sorted by effect size and plotted as a curve. Third, inferential tests:

Test 1: Median β across all specs vs. permutation null Test 2: % specs significant vs. chance expectation

Both tests use permutation: the outcome variable is randomly shuffled many times, the full specification curve is estimated for each shuffle, and the observed curve statistics are compared to the null distribution.

A finding is considered robust if (a) the median effect size across all specifications is significantly larger than the permutation null, and (b) the proportion of significant specifications is significantly higher than expected by chance.

Example Application

Orben and Przybylski (2019) applied SCA to whether screen time harms adolescent wellbeing. Using three large longitudinal datasets, they enumerated 43,890 specifications arising from different wellbeing outcomes, screen time measures, covariate sets, and subsamples. The specification curve showed that the vast majority of effect sizes were negative but extremely small — comparable in magnitude to wearing glasses. The inferential tests confirmed that while the association was real, it was far too small to support public discourse claims. The SCA replaced a single contested estimate with a complete picture of the effect landscape.

Requirements

Raw data	Required. SCA re-estimates the model under every specification from scratch.
Reproducible code	Strongly recommended. The <code>specr</code> R package and <code>rdanalysis</code> automate specification enumeration and curve estimation.
Pre-specification	Most credible when the universe of valid specifications is defined before looking at results — otherwise the choice of 'valid' specs can itself become a researcher degree of freedom.
Sample size / spec count	The permutation test gains power with either a large sample or many specifications. With very few specs and a small sample the test may have low power even when a genuine effect exists.

Tool C · **Vibration of Effects (VoE)**

Quantifying sensitivity of epidemiological associations to covariate perturbations

Overview

Vibration of effects, introduced by Chirag Patel and colleagues in 2015, addresses a specific and pervasive form of analytical fragility in epidemiology and clinical research: the sensitivity of a reported association to the choice of adjustment covariates. In fields where researchers routinely control for 'relevant confounders', the choice of which confounders to include is often undisclosed and arbitrary — and it can dramatically change both the magnitude and direction of the reported association.

Methodological Overview

The analyst begins with a set of k candidate covariates. VoE fits the regression model for every possible subset of those covariates, producing 2^k models in total. For each model, the effect size estimate and its confidence interval are recorded. The resulting distribution is summarized by three quantities:

VoE ratio = $\max(|\beta|) / \min(|\beta|)$ across all covariate subsets
Sign instability = % of models where β changes sign
CI overlap = % of models where CI excludes zero

A VoE ratio near 1 indicates a robust association. A ratio > 2 indicates substantial covariate sensitivity. Sign instability > 0% is particularly alarming — it means the direction of the association depends on which covariates are included.

Example Application

Patel et al. applied VoE to 13 associations between common exposures and health outcomes published in high-impact journals. Most associations showed substantial vibration — some changed sign depending on covariate choice, and many had VoE ratios above 2. The authors concluded that many epidemiological findings are better understood as covariate-dependent rather than reflecting stable causal relationships. The coffee-and-mortality literature is a classic example: studies that control for smoking tend to find a protective effect of coffee; those that do not tend to find null or harmful effects. VoE quantifies this instability formally.

Requirements

Individual-level dataset	Required. VoE needs access to raw data to re-fit regression models with different covariate subsets. Published summary statistics are insufficient.
Candidate covariate list	A pre-defined set of plausible confounders. Most credible when the candidate set is specified a priori or drawn from a domain consensus list.
Computational resources	For k candidate covariates, VoE fits 2^k models. For $k = 10$ this is 1,024 models — manageable. For $k = 20$ it is over one million, requiring efficient computing. The <code>voe</code> R package handles this.
Consistent outcome & exposure	All models must use the same outcome variable and exposure variable; only the covariate set varies. If outcome or exposure also vary, the tool shades into multiverse territory.

6. Shared Limitations

- All three tools require access to individual-level raw data. They cannot be applied post-publication without data sharing by the original authors — making them primarily useful as prospective methods or in re-analysis contexts where data are available.
- The definition of 'valid' or 'defensible' specifications is itself a researcher choice. Pre-registration of the specification universe is the only guard against this becoming a new researcher degree of freedom.
- A finding that is fragile across the specification space is not necessarily wrong. There may be theoretical reasons to prefer one specification over others. These tools diagnose instability; they do not determine the truth.
- Computational burden scales quickly. Multiverse and SCA become expensive for large datasets with many choice nodes; VoE becomes expensive for large covariate sets. Published R packages (`multiverse`, `sprc`, `voe`) mitigate but do not eliminate this.
- These tools are not primarily fraud-detection tools — they detect fragility, which is usually the product of normal researcher behavior rather than misconduct. Their primary value is epistemic: they tell us how confident we should be in a finding.

7. Related Tools

p-curve	Tests whether a literature shows genuine evidential value; can detect p-hacking at the study-set level but does not reveal which analytic choices produced the significant results.
z-curve 2.0	Estimates the expected replication rate from published z-values; complements specification curve by quantifying how much of a literature's signal is likely to survive replication.
SPRITE / GRIM	Operate at the single-study level on reported summary statistics; complementary to the three tools here, which operate on the raw analytic pipeline.
Preregistration	The prospective counterpart: committing to a single specification before data collection removes researcher degrees of freedom at the source rather than diagnosing them after the fact.

Key Citations

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